

Distinctive Review Sentence Extraction via Multi-Product Relative Comparison

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Abstract. This paper proposes a method for ranking review sentences by their distinctiveness to a target product. Given a target product and multiple comparison products, our approach extracts review sentences that are likely to be associated with the target product but unlikely to be associated with the comparison products. To identify review sentences that characterize a target product, we train a classifier to estimate the probability that a review sentence is associated with a product. By applying the classifier to both the target product and multiple comparison products, our method computes a target-specificity score that reflects how strongly the sentence is associated with the target product relative to the comparison products. To assess the effectiveness of the proposed approach, we conducted a user study using real-world product reviews. The results indicate that the proposed method effectively ranks review sentences that describe characteristics distinctive to the target product relative to competing products.

Keywords: Product Comparison Support · Consumer Decision Support
· Comparative Review Extraction

1 Introduction

Choosing the right product from a large number of alternatives has become increasingly difficult. Modern e-commerce platforms provide access to a vast number of products across many categories, allowing users to easily discover and compare products before making a purchase decision [19]. While this abundance of choice benefits consumers, it also increases the effort required to identify products that best match their needs. Product reviews are particularly useful in this process because they capture the experiences and opinions of previous customers. However, popular products often accumulate hundreds or even thousands of reviews. As a result, comparing multiple products by reading their reviews requires substantial time and effort [11].

Some e-commerce websites provide review summaries for individual products to reduce the burden of reading reviews. These summaries are useful for understanding what is frequently mentioned about each product. However, frequently